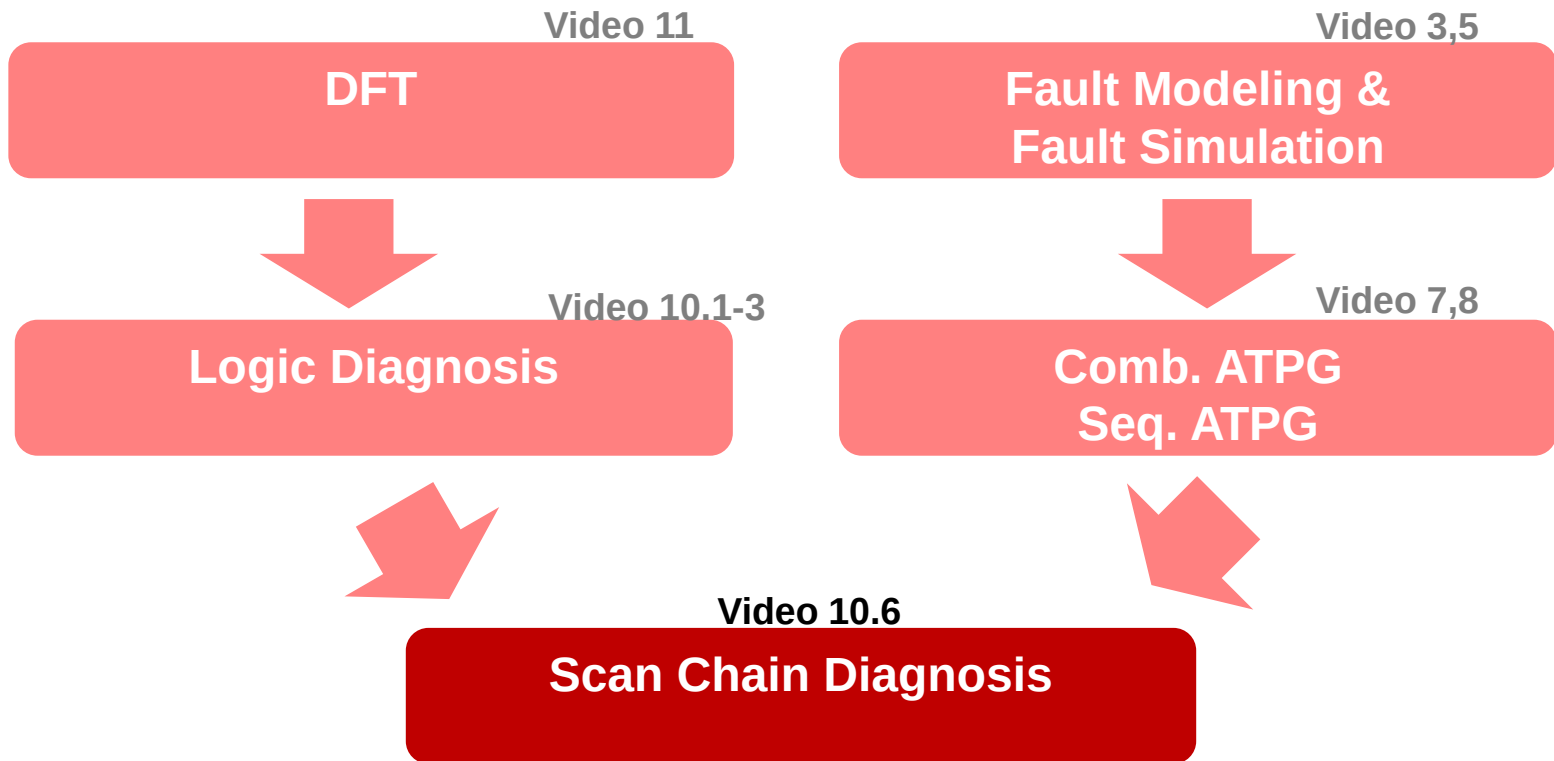


Diagnosis

- Introduction
- Logic Diagnosis
- Scan Chain Diagnosis
 - ◆ Fault models
 - ◆ Sequential ATPG diagnosis [Kundu 93]
 - ◆ Simulation-based diagnosis
 - * X simulation [Guo 01]
 - * Single excitation pattern [Li 05]
- Failure Analysis
- Conclusions

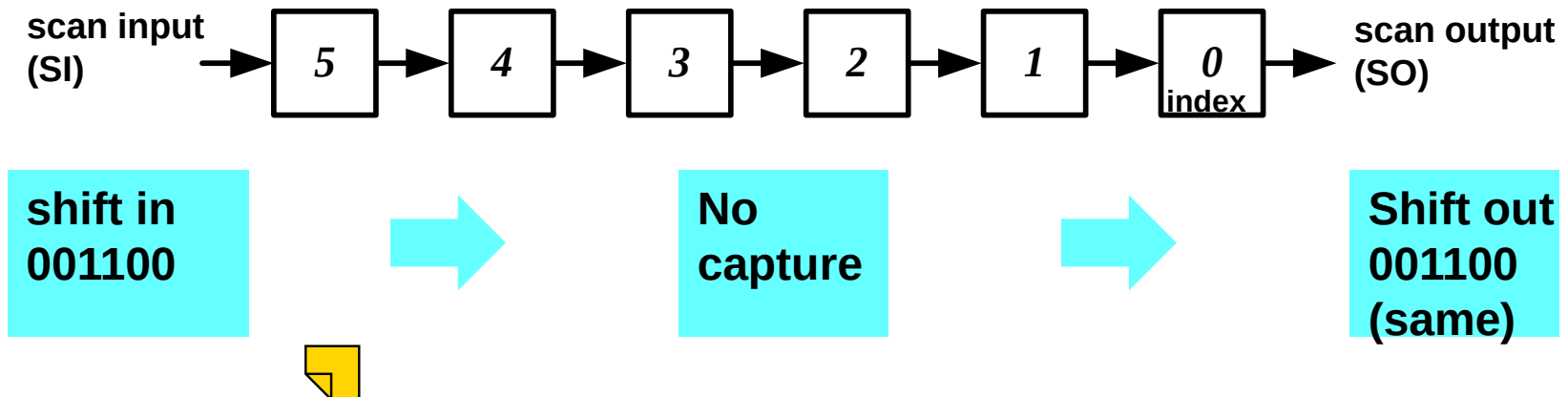


Course Roadmap for This Video



Testing Scan Chains

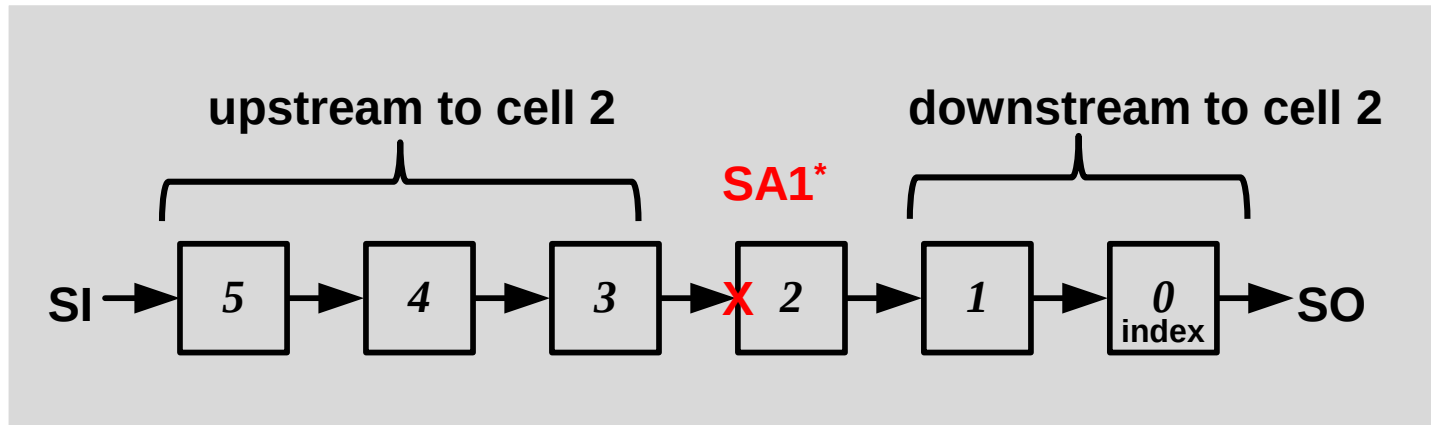
- Scan chains occupies **5%~10%** silicon area
 - ♦ **~30%** scan chain failure reported [Guo 01]
- Testing (and diagnosis) of scan chains is important
- **Scan chain integrity test (aka flush test)** applied before testing CUT
 - ♦ **Shift in 00110011...** alternating test patterns
 - ♦ **NO capture.** Directly shift out. Expect same outputs
 - ♦ All four transitions are covered $1 \rightarrow 1$; $1 \rightarrow 0$; $0 \rightarrow 1$; $0 \rightarrow 0$



Flush Test Needed for Scan Chains

Scan Chain Fault Effects

- Scan cells are indexed from SO to SI, in increasing order
- Assume cell 2 is stuck-at one
 - ♦ cells 1,2 are **downstream** to cell 2
 - ♦ cells 3,4,5 are **upstream** to cell 2



before shift in
001100

after shift in
001111

after shift out
111111

all **downstream** 0's
are flipped to one

all **upstream** 0's
are flipped to one

*NOTE: this fault located at
scan input, not data input, of cell 2

Scan Chain Fault Models

- Stuck-at one (**SA1**)
- Stuck-at zero (**SA0**)
- Slow-to-Rise (**STR**)
 - ♦ one cycle later
- Slow-to-Fall (**STF**)
 - ♦ one cycle later
- Setup Time (**ST**)
 - ♦ STR+STF
- Fast-to-rise (**FTR**)
 - ♦ one cycle earlier
- Fast-to-Fall (**FTF**)
 - ♦ on cycle earlier
- Hold-time (**HT**)
 - ♦ FTR+FTF

	flush test outputs*
Expected SO	11001100
Faulty SO (SA1)	11 <u>11</u> 11 <u>11</u>
Faulty SO (SA0)	<u>00</u> 00 <u>00</u> 00
Faulty SO (STR)	1 <u>0</u> 001 <u>0</u> 00
Faulty SO (STF)	110 <u>1</u> 1100
Faulty SO (ST)	1 <u>0</u> 0 <u>1</u> 1 <u>0</u> 00
Faulty SO (FTR)	11 <u>1</u> 011 <u>1</u> 0
Faulty SO (FTF)	1100 <u>0</u> 100
Faulty SO (HT)	11 <u>1</u> 0 <u>0</u> 1 <u>1</u> 0

* assume single **permanent fault**,
which always happens consistently

Quiz

Q: We apply flush test to a scan chain. Please identify which fault is this?

(Hint: *Intermittent fault* means a fault not always happen)

ANS:

	flush test outputs
Expected SO	11001100
Observed SO	1100 <u>0</u> 1 <u>1</u> 0

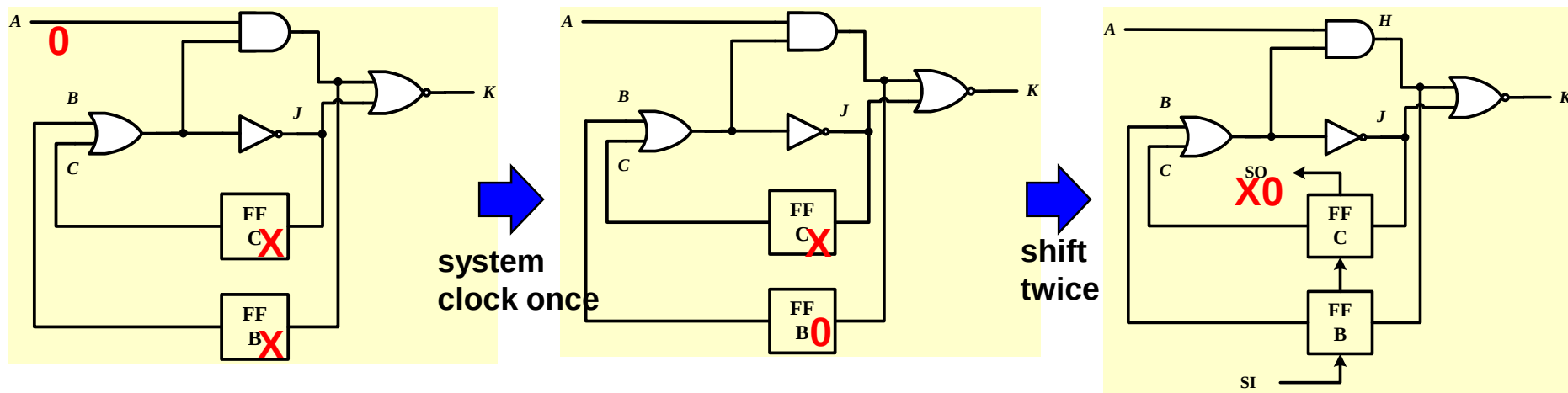
Diagnosis

- Introduction
- Logic Diagnosis
- Scan Chain Diagnosis
 - ◆ Fault models
 - ◆ Sequential ATPG diagnosis [Kundu 93]
 - ◆ Simulation-based diagnosis
 - * X-simulation [Guo 01]
 - * Single excitation pattern [Li 05]
- Failure Analysis
- Conclusions



Sequential ATPG [Kundu 93]

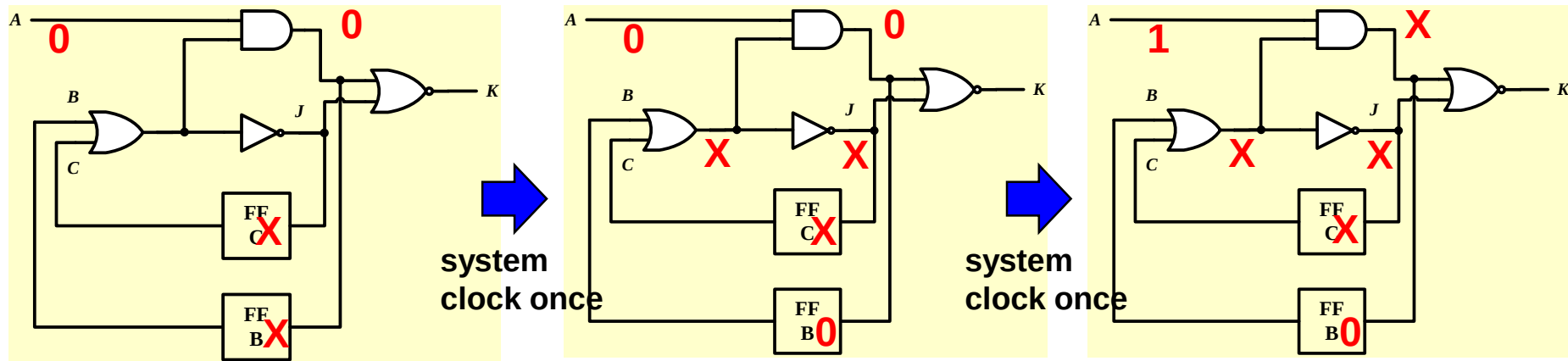
- Idea: use PI, not scan, to control FF
 - ① control FF to desired value by sequential ATPG
 - ② shift out scan chain and check its value
- Example: FF_B SA1
 - ♦ $A=0$, system clock once. Shift out 2 cycles
 - ♦ If $FF_B=0$, there is **NO SA1** downstream to FF_B ;
 - ♦ If $FF_B=1$, there is **SA1** downstream to FF_B . After shifting, the result is X1.



Repeat this to Every Cell in Chain to Locate Fault

Sequential ATPG (2)

- Another example: FF_B SA0
 - ♦ Fail to generate patterns
 - ♦ Unable to diagnose SA0 downstream to FF_B



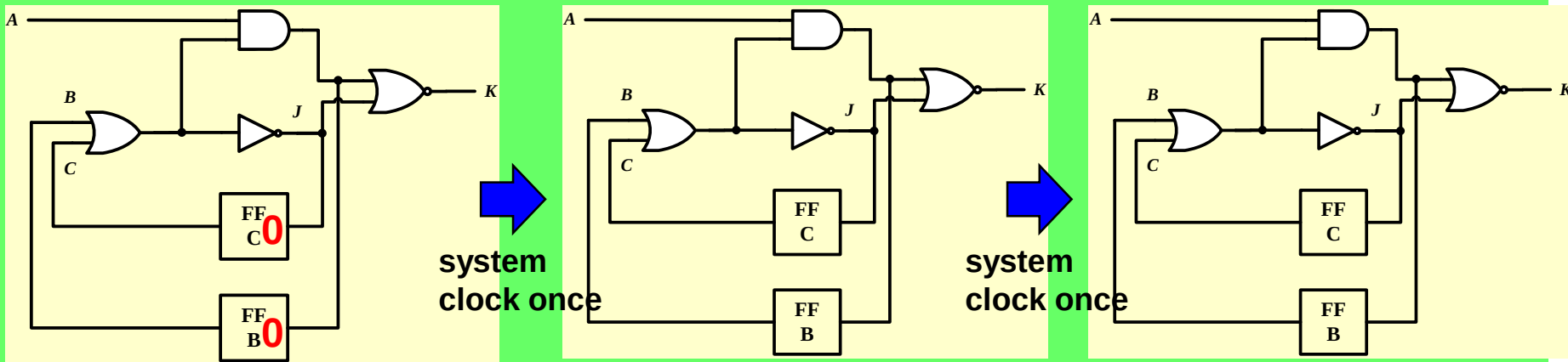
Sequential ATPG is Ineffective

Quiz

Q: John has a smart idea. Since we know there is no SA1 in scan chain, we can first shift in two zeros into the chain.

Given the initial state 00, use sequential ATPG to diagnose if there is SA0 downstream to FF_B? (that is, control FF_B = one)

ANS:



Diagnosis

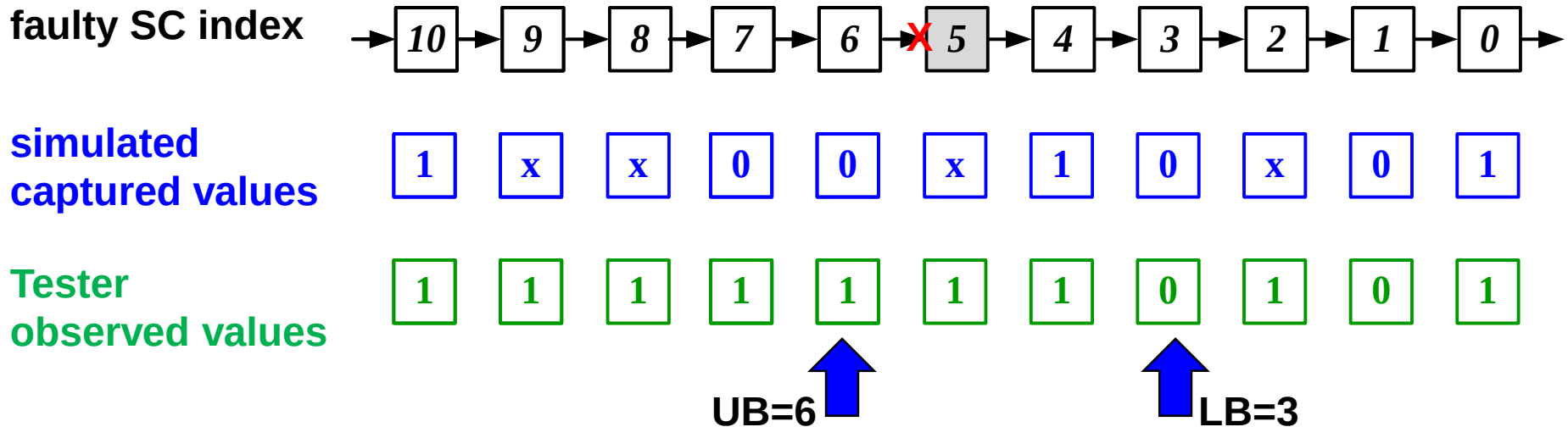
- Introduction
- Logic Diagnosis
- Scan Chain Diagnosis
 - ◆ Fault models
 - ◆ Sequential ATPG diagnosis [Kundu 93]
 - ◆ Simulation-based diagnosis
 - * X simulation [Guo 01]
 - * Single Excitation pattern [Li 05]
- Failure Analysis
- Conclusions



UB/LB by Simulation [Guo 01]

- Faulty SC and fault model is already known by flush test
- Simulate circuit with faulty SC values changed to X
 - ♦ to obtain **simulated values**
- Test CUT to obtain **observed values**
- **Lower bound** = max-index where obs. = sim. values $\neq$ stuck values
 - ♦ NOTE: X NOT considered
- **Upper bound** = mini-index where obs. $\neq$ sim. values

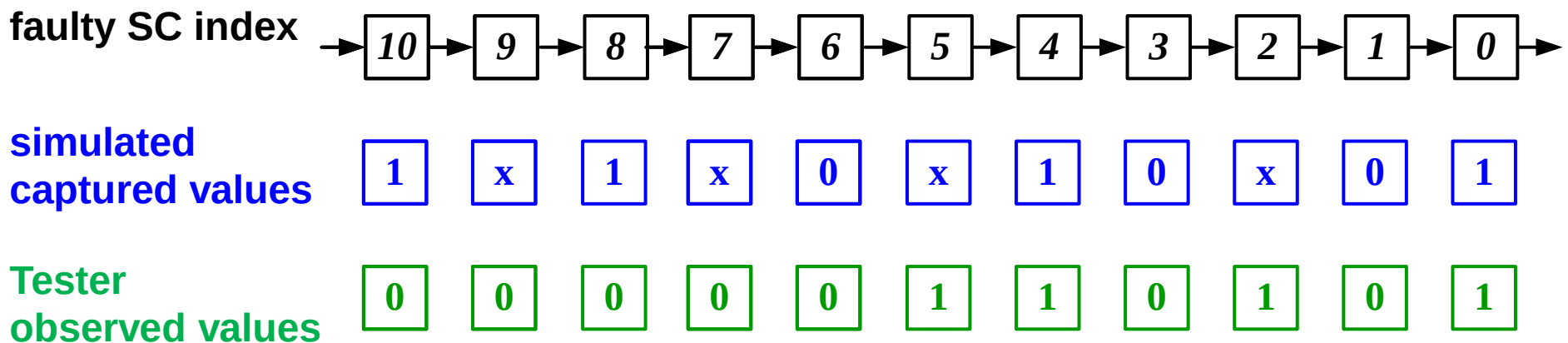
SA1



Quiz

Q: Given a SA0 fault. Given the following simulated output and observed output, what are the UB and LB?

ANS:



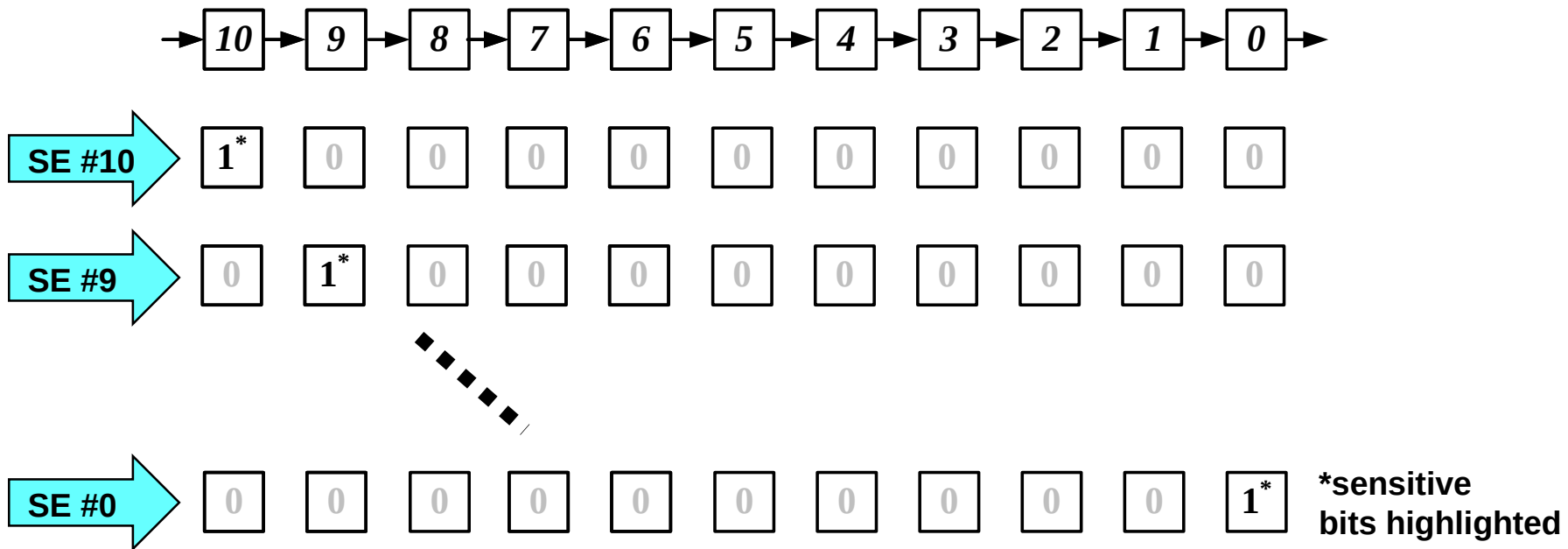
Diagnosis

- Introduction
- Logic Diagnosis
- Scan Chain Diagnosis
 - ◆ Fault models
 - ◆ Sequential ATPG diagnosis [Kundu 93]
 - ◆ Simulation-based diagnosis
 - * X simulation [Guo 01]
 - * Single Excitation pattern [Li 05]
- Failure Analysis
- Conclusions



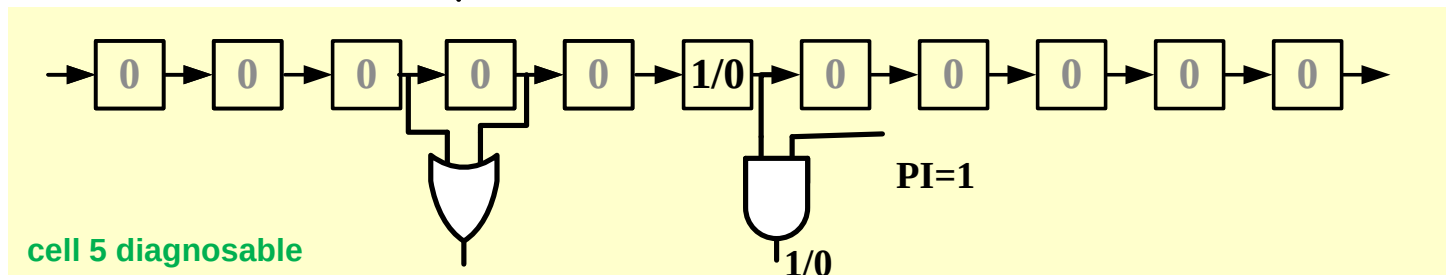
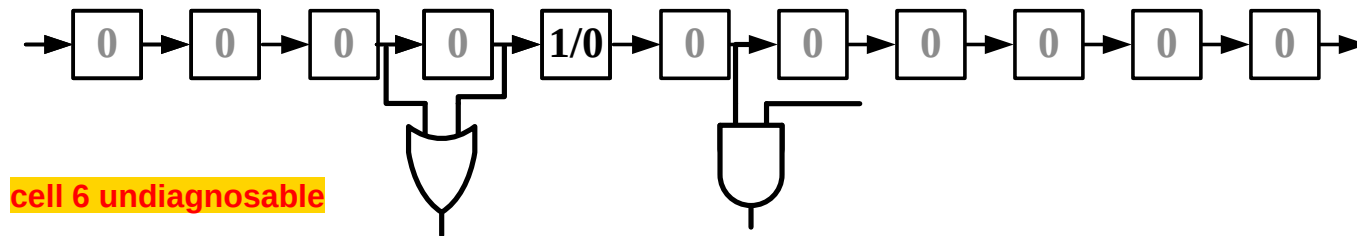
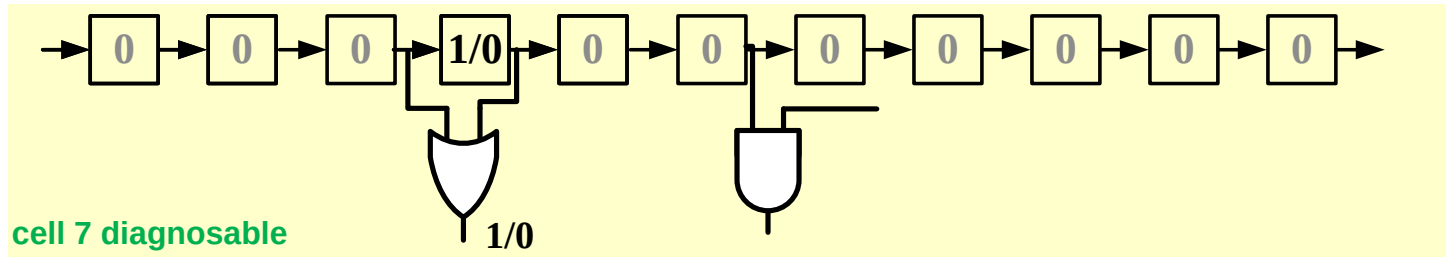
Single-Excitation Patterns [Li 05]

- Assume: faulty SC and fault model is already known by flush test
- Load **single-excitation (SE)** patterns to faulty SC
 - ♦ SE pattern has only one **sensitive-bit** can be flipped by fault
 - ♦ Generate test pattern to detect SA fault at sensitive bit
- Example: SE patterns for **SA0**



Turn Diagnosis into SSF ATPG Problem

Diagnosis Resolution



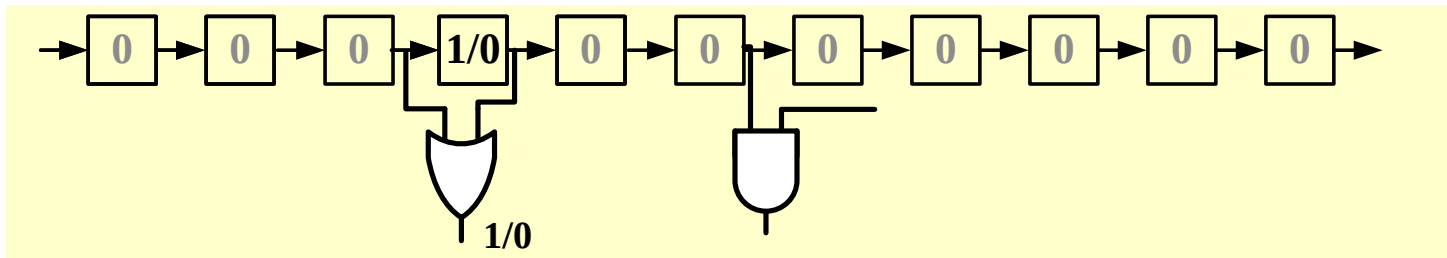
$$\text{DiagnosisResolution} = \frac{\text{total number of cells}}{\text{number of diagnosable cells}} = \frac{11}{5} = 2.2$$

Summary

- Scan chains occupies 5%~10% silicon area
 - ♦ **Scan chain faults** occur in scan chains, not logic
 - ♦ Testing (and diagnosis) of scan chain faults is important
- **Scan chain integrity test (flush test)** Used to determine the fault model and the faulty SC.
 - ♦ Shift in **00110011...** alternating test patterns. No capture
- Diagnosis techniques
 - ① Sequential ATPG (Not practical)
 - ② Simulation-based
 - * X simulation: determine **UB** and **LB** (useful for large ckt)
 - * **Single excitation** pattern (solvable by SSF ATPG)

FFT

- Q1: Since scan chain faults do not effect logic operation, why bother to test them?
- Q2: If scan cell has SA fault, can we detect by regular logic testing?
 - ♦ why bother to generate single excitation pattern?
 - ♦ e.g. this fault can be detected by regular ATPG for OR gate



Quiz

Q: Given the same circuit. Use Sequential ATPG, to diagnose if there is SA0 downstream to FF₀?

ANS:

